GIDEON LAROSE

SIMULATION ENGINEER

Winter Park FL,
32792 • 801-404-2631 • finnagainlarose@gmail.com • https://www.linkedin.com/in/gideon-larose-59284b229/

PROFESSIONAL SUMMARY

Simulation and embedded systems engineer with experience developing real-time software, XR training systems, custom operating-system components, Linux platforms, and electromechanical simulation hardware. Skilled in C/C++, Python, Unreal Engine, networking, hardware/software integration, system testing, and technical troubleshooting.

SKILLS

- **Languages:** C++, C, C#, Python, Bash, SQL, x86 Assembly
- **Platforms:** Unreal Engine 5, Unity, Meta Quest 3, Linux, FreeRTOS, Raspberry Pi, ESP32
- **Systems:** REST, MQTT, TCP/IP, GPIO, I²C, client/server architecture, Git
- **Engineering:** MOOG motion systems, CNC machining, composites, 3D printing, CAD, system calibration

ACADEMIC EXPERIENCE

FORMULA ONE MOTION BASE SIMULATOR June 2026 - Present

Final Project, Full Sail University

- Collaborated on the development and restoration of a Formula One driving simulator built around a MOOG motion-base system.
- Combined Unreal Engine simulation output with a 6-DOF MOOG motion platform, synchronizing real-time vehicle physics with industrial motion hardware.
- Tested communication between the 6-DOF simulation software, motion-control system, 3-Projector display hardware, and projection pipeline.

OS DEVELOPMENT January 2023 - Present

Personal Project, Remote

- Developing a custom x86 operating system in C and x86 assembly featuring a modular kernel architecture, multitasking, virtual memory, and low-level hardware interaction.
- Built a custom x86 boot pipeline incorporating five foundational boot subsystems, including descriptor tables, block-device drivers, initrd support, and an early graphical display abstraction.
- Construct and test components in Linux-based virtualized environments using Git and the Limine bootloader.

AI ASSISTANT March 2022 - Present

Personal Project, Remote

- Developed and fine-tuned a locally hosted language model using custom datasets and LoRA workflows for domain-specific conversational behavior.
- Developed Python-based inference, speech recognition, text-to-speech, SQLite persistence, and contextual retrieval services for offline operation.

JUSTINTIME May 2026 - July 2026

Project and Portfolio Courses, Full Sail University

- Constructed an Unreal Engine nursing-training application integrating five AI subsystems: speech-to-text, local LLM inference, text-to-speech, persistent conversational context, and contextual media retrieval.
- Extended Unreal Engine plugins and established client/server communication for voice streaming, video delivery, and AI requests between XR clients and Linux services.

EDUCATION

Bachelor of Science in Simulation & Visualization (B.S.)• Full Sail University, Winter Park, FL
August 2026

Relevant coursework: C++, computer graphics, simulation, XR development, embedded systems, networking, systems integration, and hardware-focused engineering projects.